

LearnPro AI Interview Preparation Platform

Project Overview

LearnPro is an AI-powered interview preparation and career guidance platform designed for students, job seekers, and professionals. The platform combines learning resources, career roadmaps, mock interviews, subject-wise tests, analytics, and an AI mentor chatbot to help users prepare for technical interviews and achieve their career goals.

The platform provides:

- Career Roadmaps
 - Subject-wise Resources
 - Mock Interviews
 - Interview Question Practice
 - AI Interview Mentor
 - Learning Analytics
 - Progress Tracking
 - Personalized Recommendations
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User Roles

The platform supports the following user categories:

1. Student
2. Job Seeker
3. Professional

The platform also supports career-specific learning paths:

- Web Developer
 - Frontend Developer
 - Backend Developer
 - Full Stack Developer
 - Software Development Engineer (SDE)
 - AI Engineer
 - Machine Learning Engineer
 - Data Scientist
 - Data Analyst
 - Cloud Engineer
 - DevOps Engineer
 - Cyber Security Engineer
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Platform Workflow

Step 1: Landing Page

Route: /

The landing page introduces LearnPro and its features.

User actions:

- Get Started
- Login
- Explore Features

If the user clicks Get Started: → Onboarding Page

If the user clicks Login: → Sign In Page

Step 2: Onboarding

Route: /onboarding

The onboarding page allows users to select their learning goal.

Available paths:

- Student
- Job Seeker
- Professional

The selected role is stored for personalization.

Example: Role = Student

The platform will recommend:

- Learning resources
- Roadmaps
- Mock interviews
- Practice tests

based on the selected role.

Step 3: Authentication

Routes:

- /signin
- /signup

Available methods:

Google Login

The platform can collect:

- Name
- Email
- Profile Picture
- Google ID

OTP Login

Workflow:

Enter Email → Generate OTP → Send OTP to Email → Verify OTP → Login Success

Manual Registration

User provides:

- Full Name
- Email
- Phone Number
- Password
- Career Role

Data is stored in the database.

Step 4: Dashboard

Route: /dashboard

The dashboard is the central hub of the platform.

Displayed Information:

Learning Statistics

- Average Test Accuracy
- Weekly Study Hours
- Courses Completed

- Resource Completion Rate

Subject Mastery

Tracks proficiency in:

- DSA
- DBMS
- Operating System
- Computer Networks
- OOPS
- Frontend
- Backend
- AI
- Machine Learning
- System Design

Weekly Activity

Displays:

- Study Hours
- Tests Completed
- Resources Visited

Strengths

Shows top-performing subjects.

Example:

- Algorithms
- DBMS
- OOPS

Improvement Areas

Shows weak subjects.

Example:

- System Design
- Operating System
- Computer Networks

Resources Section

Route: /library

Purpose: Provide learning materials.

Subjects Available:

- DSA
- DBMS
- Operating System
- Computer Networks
- OOPS
- C Programming
- C++
- Java
- Python
- JavaScript
- SQL
- Machine Learning
- Deep Learning
- Artificial Intelligence
- NLP
- Computer Vision
- Generative AI
- Cloud Computing
- System Design
- Software Engineering

Each subject contains:

YouTube Playlists

Examples:

- Striver
- Apna College
- Love Babbar
- CodeHelp
- Gate Smashers
- FreeCodeCamp

Websites

Examples:

- GeeksforGeeks
- LeetCode
- InterviewBit
- W3Schools
- MDN
- Scaler

Users can:

- Open resources

- Bookmark resources
 - Mark resources as completed
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Roadmaps Section

Route: /roadmaps

Purpose: Provide structured career guidance.

Available Roadmaps:

Web Development

HTML → CSS → JavaScript → React → Node.js → Express → MongoDB → Projects → Placement Preparation

AI Engineer

Python → Mathematics → Statistics → Machine Learning → Deep Learning → NLP → Computer Vision → LLMs → RAG → AI Projects

Data Scientist

Python → SQL → Statistics → Data Analysis → Machine Learning → Data Visualization → Projects

SDE

Programming → DSA → DBMS → Operating System → Computer Networks → System Design → Projects → Interview Preparation

Subject Test Section

Route: /exam

Purpose: Practice interview questions.

Available Subjects:

- DSA
- DBMS
- OOPS
- Operating System
- Computer Networks
- Java
- Python
- SQL

- Machine Learning
- Deep Learning
- Artificial Intelligence

Features:

- MCQ Questions
- Previous Question Navigation
- Flag Questions
- Skip Questions
- Instant Feedback
- Score Tracking

After Submission:

User receives:

- Correct Answer
 - Wrong Answer
 - Score
 - Performance Analysis
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Mock Interview Section

Route: /interview

Purpose: Simulate real interviews using AI.

Interview Types:

- Web Development
- SDE
- AI Engineer
- Machine Learning Engineer
- Data Scientist
- HR Round

Workflow:

AI Avatar appears.

AI asks:

"Tell me about yourself."

User answers.

AI evaluates:

Technical Accuracy

Checks whether the answer is technically correct.

Completeness

Checks missing concepts.

Communication Skills

Evaluates:

- Clarity
- Confidence
- Fluency

Problem Solving

Evaluates:

- Logic
- Approach
- Explanation

AI then provides:

Score Strengths Weaknesses Improvement Suggestions

Next Question is generated automatically.

At the end:

Final Interview Report includes:

- Technical Score
- Communication Score
- Confidence Score
- Overall Score
- Recommended Resources
- Suggested Roadmaps

AI Interview Mentor

Route: /chat

Purpose: Provide intelligent guidance using RAG.

The chatbot knows:

- Platform Navigation
 - Resources
 - Roadmaps
 - Interview Questions
 - Learning Paths
 - Career Guidance
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Platform Navigation Knowledge

If a user asks:

"I want to practice DBMS."

Response:

Navigate to: Exam Page → Select DBMS

If a user asks:

"I need DBMS resources."

Response:

Navigate to: Library → DBMS

If a user asks:

"I want to become an AI Engineer."

Response:

Navigate to: Roadmaps → AI Engineer Roadmap

If a user asks:

"I want to give a mock interview."

Response:

Navigate to: Mock Interview Page

Route: /interview

If a user asks:

"How can I track my progress?"

Response:

Navigate to: Dashboard

Route: /dashboard

If a user asks:

"I forgot where the learning materials are."

Response:

Navigate to: Library

Route: /library

AI Career Guidance Knowledge

AI Engineer Skills

Required Skills:

- Python
 - Mathematics
 - Statistics
 - Machine Learning
 - Deep Learning
 - NLP
 - Computer Vision
 - Generative AI
 - RAG
 - LLMs
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Data Scientist Skills

Required Skills:

- Python
- SQL
- Statistics

- Data Cleaning
 - Data Visualization
 - Machine Learning
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SDE Skills

Required Skills:

- Programming
 - DSA
 - DBMS
 - Operating System
 - Computer Networks
 - System Design
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Web Developer Skills

Required Skills:

- HTML
 - CSS
 - JavaScript
 - React
 - Node.js
 - Databases
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Frequently Asked Questions

Q: How do I start learning AI?

A: Go to Roadmaps → AI Engineer.

Q: How do I practice interview questions?

A: Go to Exam Page and select a subject.

Q: How do I take a mock interview?

A: Navigate to Mock Interview.

Q: How do I access resources?

A: Open Library and choose a subject.

Q: How do I check my performance?

A: Open Dashboard.

Q: How can I improve weak subjects?

A: Review Improvement Areas in Dashboard and study recommended resources.

Goal of LearnPro

The primary goal of LearnPro is to provide a complete AI-powered ecosystem for:

- Learning
- Career Planning
- Interview Preparation
- Skill Assessment
- Progress Tracking

through a unified platform driven by intelligent recommendations and AI mentoring.